

DILSHAN PATHEGAMAGE

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building web and mobile applications using the MERN stack, Flutter, and ASP.NET Core. Experienced in applying these technologies to develop RESTful APIs, working with relational and non-relational databases, integrating AI into practical solutions, and working with cloud and DevOps technologies. I combine practical problem-solving with adaptability across technologies to build reliable, maintainable solutions that address real-world needs and contribute to product and team success.

WORK EXPERIENCE

Trainee Software Engineer at Virstack Pvt Ltd

DEC 2025 - PRESENT

Tech stack: MERN, Flutter/Dart, Docker, AWS, NGINX, CI/CD, WebSockets

- Developed end-to-end project modules, contributing to planning, ER design, development, and production release with senior developers.
- Designed and optimized scalable RESTful APIs and modular application architectures, improving performance and long-term maintainability.
- Built a Bluetooth GATT server enabling data transfer between mobile devices and VR glasses, and worked with WebSockets for real-time communication.
- Contributed to AI-integrated solutions and managed cloud deployments, TestFlight releases, App Store submissions, and DevOps workflows with Docker, AWS, NGINX, and CI/CD, streamlining releases within agile international teams.
- Wrote unit and integration tests to ensure code quality and reliability, validating features before production release.

Intern Software Engineer at Virstack Pvt Ltd

MAR 2025 - NOV 2025

Tech stack: MERN, Flutter/Dart

- Designed and developed end-to-end features for web and mobile applications across procurement, healthcare, meditation, and AI-driven platforms, including database and full-stack development.
- Engineered optimized RESTful APIs and MongoDB aggregation pipelines for search, filtering, scheduling, and data-driven workflows, improving data retrieval efficiency and application responsiveness.
- Developed end-to-end features including authentication, authorization, user and profile management, scheduling, booking, content management, notifications, and third-party integrations, expanding platform functionality and improving user workflows.
- Performed root cause analysis and resolved production issues across web and mobile applications, improving stability and user experience.

EDUCATION

Bsc.(Hons) in Information Technology - University of Moratuwa

JUN 2022 - JUL 2026

- Level 2 Semester 2 SGPA : **3.90 - Dean's List**
- Level 4 Semester 1 SGPA : **3.88 - Dean's List**
- Overall GPA - **3.55** (out of 4.0 - up to L4S1)

G.C.E. Advanced Level - Galahitiyawa Central College, Ganemulla

JUN 2017 - OCT 2020

- Combined Maths - A | Chemistry - B | Physics - B

TECHNICAL SKILLS

Programming Languages :	Java, JavaScript, TypeScript, Dart, C#, Python
Frontend:	React.js, Flutter(Mobile App Development), Redux, Tailwind CSS, Bootstrap, HTML, CSS
Backend:	Node.js, Express.js, ASP.NET, REST APIs, WebSockets, JWT, OAuth
Databases	MongoDB, MYSQL, MSSQL, Neo4j
Cloud & DevOps :	AWS (EC2, ECS, S3, Lightsail, CloudWatch, Lambda), Docker, NGINX, Kubernetes, CI/CD Pipelines
Software Engineering :	Scalable System Design, Secure Software Design, Agile Development, Design Patterns, Performance Optimization, OOP Principles, SOLID Principles, Version Control, Data Structures & Algorithms, Microservices, TDD, AI-assisted development
Testing & Quality Assurance :	Cypress, Cucumber, Unit Testing, End-to-End Testing, Automated Testing

SOFT SKILLS

Problem Solving, Communication, Teamwork & Collaboration, Adaptability, Time Management, Leadership, Eagerness to Learn, Motivation

PROJECTS

Final Year Research - Exploring Hardware-Software Codesign of Spatial Deep Learning Hardware Accelerators (Group)

Developed a cycle-accurate hardware-software co-design simulator framework for systolic array-based DNN accelerators, addressing critical performance bottlenecks in DNN execution. Researched and designed the compute and data delivery module of the simulator, exploring loop optimizations, memory management schemes, and dataflow strategies to improve accelerator performance. Validated against hardware, confirming cycle-accurate results.

- Role - Researcher
- Technologies - Python, System Verilog, DNN Workloads, FPGA Simulation

[↗](#) GitHub repository

Sri Lanka Tourist Arrivals Prediction Using Machine Learning

Developed a machine learning-based forecasting system to predict Sri Lanka’s monthly foreign tourist arrivals using tourism data from 2019–2024. Built and optimized a Random Forest regression model with advanced feature engineering, time-series forecasting, and SHAP explainability, achieving 92% R² prediction accuracy.

- Role - ML Engineer, Full Stack Developer
- Technologies - Python, Scikit-learn, SHAP, Machine Learning, Data Analysis

[↗](#) GitHub repository

Expert System - University Course Selection Advisor

Developed a rule-based expert system that automates university course selection by evaluating prerequisites, credit requirements, and career paths, replacing a manual, error-prone process. Built career-based recommendations, prerequisite and automated credit validation on a Prolog knowledge base, with an interactive Python GUI for decision support.

- Role - Full Stack Developer
- Technologies - Prolog, Python, Tkinter

[↗](#) GitHub repository

TicketMate - Public Transport E-Ticketing System (Level 2 Software Project - Mentored by Bistec Global)

Contributed to a public transport web application for train and bus ticketing with online payments, live tracking, and scheduling. Developed search, reservation, and cancellation/refund workflows, passenger profile management, and a review and rating system, along with the UI/UX design for these modules.

- Role - Full Stack Developer, Ui/Ux Designer
- Technologies - React Js, TypeScript, Asp.Net, MsSql, Bootstrap, Azure Cloud

[↗](#) Front-end repository

[↗](#) Back-end repository

MozzaMelt - Ecommerce Platform (Group)

Built a full-featured pizza ordering web application with payment, order tracking, reviews, and admin management. Developed the review and rating service, including backend logic and unit testing.

- Role - Full Stack Developer
- Technologies - React Js, Asp.Net Core, MsSql, Docker, Ci/Cd Pipeline, Xunit

[↗](#) Front-end repository

[↗](#) Back-end repository

ACHIEVEMENTS AND PARTICIPATIONS

- SLIIT Codefest 2022-Finalists : Merit Award recipient in the tertiary category at the final round.
- Openhack 2.0 Designathone 2023-Finalists - Organized by IIT : Merit Award recipient at the final round.
- HackX 2023 - Organized by University of Kelaniya : Semi-finalist.
- Code Rush 2023 - Intra Faculty Coding Competition : Participant.
- Devthon 2024 : Participant.
- IDEALIZE 2024 - Organized by AIESEC in University of Moratuwa : Semi-finalist.
- HackMoral 6.0 2024 Organized by INTECS : 32 nd place in the batch 21 at faculty of IT (Individual)

EXTRA- CURRICULAR ACTIVITIES

- FIT Moments : Member of the HR pillar of IT Faculty Official Media Unit.
- School cricket team (Hard-ball) : Captain (2013 - 2015) , Member (2011 - 2016)
- School Prefect : 2010, 2016
- Rotaract Club of UoM : Organizing committee member - Binara Padura 2023
- Mora Hiking Club : Member of the hiking club

REFERENCES

Mr. Yashod Perera

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Mr. Rushen Samodya

Software Engineer at Geveo.

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